QIN, YUAN

+41 766058207 | yuaqin@student.ethz.ch | qinyuan2000.github.io

EDUCATION

ETH Zurich, Dept. of Information Technology and Electrical Engineering — Zurich, Switzerland
Master of Science, Electrical Engineering and Information Technology — 09/2023 – 05/2026

- **GPA: 5.67/6.00**, approx. top 14%
- **Selected coursework**: High-Level Synthesis, Graph Theory, Discrete Event System, Mathematical Optimization, Computer Architecture, Machine Learning

Southeast University, School of Electronic Science & Engineering — Nanjing, China
Bachelor of Engineering, Electronic Science and Technology — 09/2019 – 06/2023

- **GPA: 3.94/4.00**, ranking: top 3%

RESEARCH EXPERIENCES

A Systematic Approach to Robust Fast Token Delivery (Master Thesis) — 09/2025 – 04/2026
Advisors: Prof. Lana Josipović (ETH Zurich) & Prof. Paolo Ienne (EPFL)

- Developed a complete Fast Token Delivery (FTD) procedure for dynamically scheduled dataflow circuits, covering GSA conversion, regeneration, and suppression.
- Identified and resolved a correctness gap where suppression circuits introduce internal condition-token deliveries that can violate token-matching invariants.
- Designed and proved a robust suppression algorithm that uses dominance, control dependence, control-flow subgraphs, and ROBDD-based Boolean representations to synthesize MUX/BRANCH suppression circuits while preserving internal token matching.
- Implemented the end-to-end FTD flow in the Dynamatic compiler using C++ MLIR passes, lowering control-flow IR to the Handshake dialect.
- Proposed generalized decision-diagram-to-circuit mappings beyond standard Shannon expansion to reduce redundant condition-token uses in the generated suppression logic.

Refactoring & Extending MILP-Based Buffer Placement (Master Project) — 10/2024 – 06/2025
Advisor: Prof. Lana Josipović (ETH Zurich)

- Refactored the buffer placement infrastructure within the Dynamatic compiler (C++/MLIR) to support explicit buffer-type classification, enabling diverse buffer implementations with fine-grained area-elasticity trade-offs.
- Integrated the Cost-Aware-Buffers algorithm with generalized throughput constraints and LUT-based area objectives, and resolved critical modeling bottlenecks (e.g., latency imbalances).
- Proposed a two-stage optimization strategy that linearizes quadratic constraints via precomputed throughput values, transforming the computationally expensive MIQCP formulation into an efficient MILP model while maintaining solution optimality.

Cost-Aware Buffer Placement for Dataflow Circuits (Master Project) — 02/2024 – 07/2024
Advisor: Prof. Lana Josipović (ETH Zurich)

- Formulated a MILP-based optimization model capturing elasticity and area characteristics of heterogeneous buffer types, enabling finer-grained circuit throughput and area optimization.
- Implemented the optimization framework in Python and validated it via Vivado synthesis, achieving 4.8%–25.5% area reduction on PolyBench benchmarks without throughput degradation compared to state-of-the-art methods.
- Explored a theoretical framework for buffer optimization in the presence of units with variable latency and initiation interval.

Statistical Cell Delay Modeling under Input Transition Time Variability (Bachelor Thesis) 12/2022 – 05/2023

Advisor: Prof. Peng Cao (Southeast University)

- Developed a statistical delay model for sub-threshold inverter chains under correlated threshold-voltage and input slew variations.
- Proposed a piecewise-log waveform approximation method to derive closed-form delay expressions, enabling consistent and accurate propagation modeling across inverter stages.
- Validated the statistical model on a 28 nm PDK: Monte Carlo simulations show 2.9–3.7× lower error in delay distribution prediction compared to prior methods.

Roadside Cone Detection and Tracking for Racing Track Calibration (Summer Project) 07/2022 – 09/2022

Advisor: Prof. Quoc-Viet Dang (UC Irvine, remote)

- Engineered a vision-based perception pipeline that projects 2D cone detections into vehicle-centric world coordinates via homography and vehicle dynamics modeling.
- Formulated a trajectory optimization model using Gurobi to calibrate noisy GPS data by minimizing cone-to-vehicle positional errors, ensuring high-precision track mapping.

Interdependent Flip-flop Timing Analysis & Optimization for Near-Threshold Voltage Design 08/2021 – 06/2022

Advisor: Prof. Peng Cao (Southeast University)

- Developed an ANN-based interdependent timing model to capture the nonlinear setup/hold time and clock-to-q delay relationships in Near-Threshold Voltage flip-flops, achieving <0.69% error with minimal simulation overhead.
- Integrated the model into a Static Timing Analysis flow and implemented an iterative optimization strategy that balances timing slack across critical paths, reducing the minimum clock period by 1.7-6.3% without hardware cost.
- Validated the framework on ISCAS'89 benchmarks (SMIC 40nm, 0.6V), demonstrating 6.7% performance gains with logarithmic runtime scaling, and published the findings in *Electronics* (2022).

HONORS & AWARDS

- **National Scholarship**, Ministry of Education of China (Top 0.2% Nationwide) 2021
- **Zhongnan Scholarship**, School of Electronic Science & Eng. (1 recipient in School) 2021
- **1st Prize**, Advanced Mathematics Competition at Southeast University (Top 2%) 2020
- **1st Prize**, Jiangsu Provincial Advanced Mathematics Competition (Top 3.5%) 2020
- **1st Prize**, The 12th National College Mathematics Competition (Top 7%) 2020
- **SEU Single-course Scholarships** (Top 5% in seven major courses) 2020-2021
- **Scholarship for Excellence** in Student Service (1 recipient in School) 2021

SKILLS & ADDITIONAL

Skills: C++, Python, MLIR, VHDL, Verilog, x86 Assembly, MATLAB, Gurobi, Git, LaTeX

Standardized Tests: TOEFL 107 (R30, L27, S23, W27); GRE 331 (V: 161, 87%; Q: 170, 96%)